

VINAY KRISHNA MURTHY

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EDUCATION

Northeastern University

Master of Science in Information System

(Sep 2023 - May 2025, USA)

Nitte Meenakshi Institute of Technology

Bachelor of Engineering in Electronics and Communication Engineering.

(Aug 2017 - July 2021, INDIA)

TECHNICAL SKILLS

Programming: Java, C++, Python, JavaScript, TypeScript

Frontend: React.js, Next.js, Redux, HTML5, CSS3, AJAX, REST/GraphQL APIs

Backend: Spring Boot, Node.js (Express), Hibernate JPA, Flask, FastAPI, gRPC

Databases: MySQL, PostgreSQL, Oracle, MongoDB

Cloud Platforms: AWS (EKS, ECS, EC2, Lambda, S3, RDS, CloudWatch), GCP, Azure

DevOps & Tools: Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Prometheus, Grafana

Testing & Quality: Selenium, Playwright, JMeter, Postman, CI/CD test automation

Productivity & Collaboration: MS Office (Excel, PowerPoint, Word, Outlook), JIRA, Confluence

WORK EXPERIENCE

One Community Inc(Software Engineer)

(May 2025 – Present, USA)

- Built and enhanced scalable full-stack features across **React/Redux (HighestGoodNetworkApp)** and **Node.js/Express (HGNRest)**, delivering reliable, maintainable functionality in a fast-paced Agile environment.
- Developed and optimized backend services and **REST APIs**, implementing robust validation/error handling and improving reliability for production workflows.
- Worked with **MongoDB** for data modeling and query optimization, and integrated **ElasticSearch** for efficient search/indexing flows and faster retrieval.
- Implemented **responsive UI components** and clean state management, collaborating with **UI/UX** and engineers to ship improvements while keeping changes modular and low risk.
- Managed multiple parallel tasks by scoping work, estimating effort, and updating tracking docs to ensure strong follow-through and on-time delivery.

Accenture Solutions Pvt Ltd(Software Engineer)

(Aug 2022 – Aug 2023, INDIA)

- Designed and deployed **50+ microservices** as part of **large-scale infrastructure and distributed systems** using Java and Python on **AWS EKS/ECS**, supporting **120K+ users**.
- Built and optimized **REST and gRPC APIs** processing **2M+ messages/day**, achieving sub-100ms latency and improved fault tolerance.
- Triaged and debugged **large-scale production system issues**, including concurrency, asynchronous I/O, and memory management, reducing service crashes by **30%**.
- Improved scalability via load balancing, service refactoring, and performance tuning, cutting response times by **20%**.
- Secured services using **JWT/OAuth2** and **TLS/SSL** and implemented monitoring and alerting using **Prometheus and Grafana**.
- Participated in **design reviews and code reviews**, contributing to technical documentation and operational runbooks.

Accenture Solutions Pvt Ltd(Associate Software Engineer)

(Aug 2021 - Aug 2022, INDIA)

- Designed and implemented **Spring Boot microservices** and **Flask APIs**, improving application performance by **25%**.
- Collaborated with UI/UX designers to build **React.js + Redux** front-end interfaces integrated with backend REST APIs.
- Automated **Docker-based CI/CD pipelines** on **AWS EC2**, reducing manual deployment errors by **20%**.
- Tuned **PostgreSQL and Redis queries**, improving latency by **30%** using indexing and caching strategies.
- Delivered **Agile sprint deliverables** across multi-disciplinary teams, ensuring timely releases with minimal defects.

ACADEMIC PROJECTS

Real-Time Chat Application (Vin Chat)

- Engineered a **low-latency, event-driven** real-time messaging backend using **Java (Spring Boot)**, **WebSockets**, and **RabbitMQ**, with reliable delivery and presence tracking for scalable concurrent users.

Online Learning Management System (LMS)

- Built **scalable, backend-first REST APIs** using **Java (Spring Boot)** / **Python (FastAPI)** with **role-based access**, optimized **PostgreSQL queries**, and production-ready validation/error handling for reliable course and progress workflows.